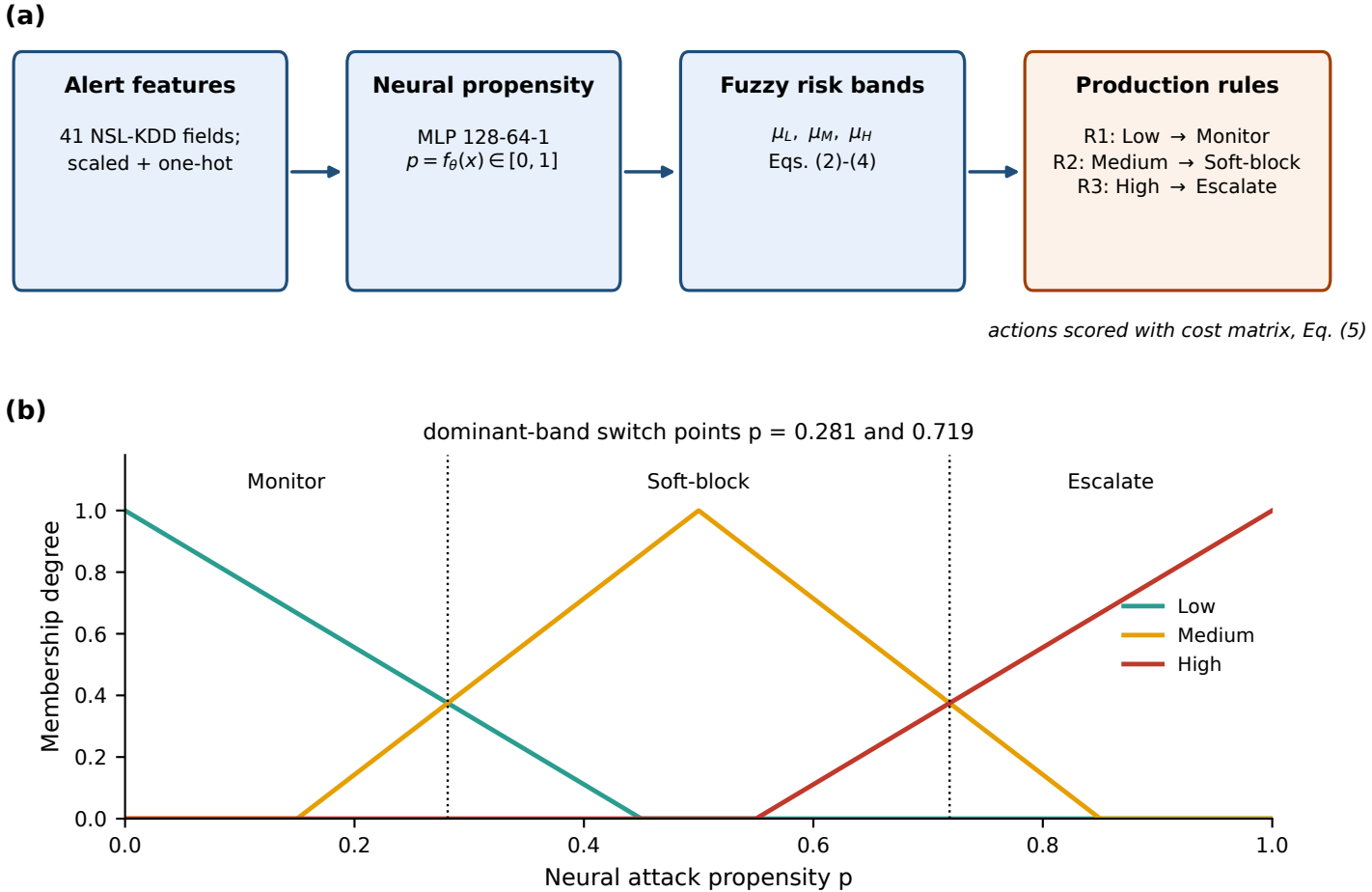
(a)
Alert features
41 NSL-KDD fields;
scaled + one-hot
Neural propensity
MLP 128-64-1
$p = f_\theta(x) \in [0, 1]$
Fuzzy risk bands
$\mu_L,\ \mu_M,\ \mu_H$
Eqs. (2)-(4)
Production rules
R1: Low → Monitor
R2: Medium → Soft-block
R3: High → Escalate
actions scored with cost matrix, Eq. (5)
(b)
dominant-band switch points p = 0.281 and 0.719
Monitor
Soft-block
Escalate
Low
Medium
High
Membership degree
Neural attack propensity p
0.0
0.2
0.4
0.6
0.8
1.0